
Atmospheric correction

Launcher: `Toolbox/atmcorr.bat`

Purpose

Atmospheric correction converts sensor digital numbers into estimated bottom-of-atmosphere surface reflectance. The current GUI applies DOS3 (Dark Object Subtraction) to both the multispectral and panchromatic products in a WorldView product folder. This should normally be the first processing step.

DOS3 estimates haze from dark pixels, applies sensor radiometric calibration and acquisition geometry, compensates approximately for Rayleigh scattering, and clips negative reflectance to zero. It is an empirical correction, not a full atmospheric radiative-transfer model.

Inputs and parameters

Option	What to provide	Behaviour
Input Image Folder	The top-level directory of one WV3 or LEGION-06 product	It must contain one folder ending <code>_MUL</code> and one ending <code>_PAN</code> . Those folders must retain their <code>.TIF</code> , <code>.TIL</code> , <code>.IMD</code> , and <code>.XML</code> product metadata; the parent product folder must retain its <code>*README.TXT</code> . The first matching <code>MUL</code> and <code>PAN</code> folders are used.
Output Folder	An existing folder, or blank	Blank creates <code><input-folder>/<input-folder-name>_BOA</code> . If that automatic folder already exists, the GUI asks you to choose another folder or rerun. A manually entered folder is expected to exist.

Option	What to provide	Behaviour
Vector layer	Optional clip polygon	The raw product is cropped before correction. Pixels outside the cutline become NoData –9999. Leave blank to process the full scenes.
Sensor	WV3 or LEGION_06	Select the sensor that produced the imagery. It controls calibration coefficients and metadata interpretation. Default: WV3.
Atmospheric Correction formula	DOS3	DOS3 is the currently available formula and the default.
Scale	A non-negative integer	0 keeps floating-point reflectance, normally around 0–1. A positive value multiplies reflectance and writes signed 16-bit integers. 10000 is recommended for the conventional Bayesian pansharpener workflow. Negative and non-integer values are rejected.

Outputs

For each detected MUL and PAN image, the tool creates:

- <source-name>_DOS3.tif

Outputs are tiled, DEFLATE-compressed GeoTIFFs with NoData –9999 and overviews. The source data are not modified. Temporary clipped files are removed after a successful run.

Choosing a scale

- Use 10000 when the next step is conventional Bayesian pansharpening. It preserves four decimal places while giving the solver image-like numeric magnitudes.
- Use 0 when another scientific workflow specifically expects floating-point reflectance. If Bayesian pansharpening follows, start with a much lower lambda (around 0.1) and inspect band statistics carefully.
- Because positive scaling writes `int16`, values above about 3.2767 at scale 10000 would clip to the data-type maximum. Normal reflectance usually stays below this limit.

Quality checks

Open both outputs and verify their CRS, footprint, band count, and NoData border. Inspect per-band histograms: most valid reflectance should be non-negative, and multispectral bands should retain distinct distributions. Strong haze, cloud, water, shadow, snow, or an unrepresentative dark-pixel estimate can bias DOS3.